

MACHINE TOOL TECHNOLOGY

WEEK-5

MECHANICAL 3RD SEMESTER

LATHE MACHINE TOOL OPERATION

The various operations that can be done with the help of lathe machine are as follows:-

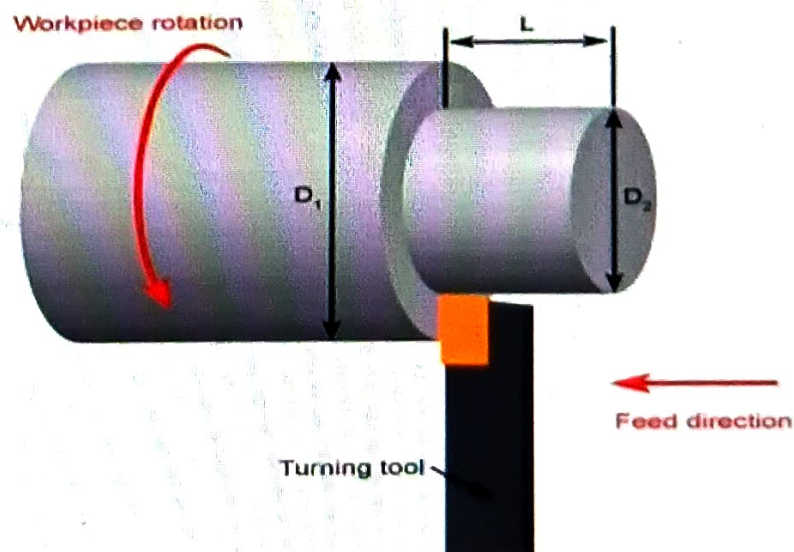
1. Turning
2. Facing
3. Chamfering
4. Knurling
5. Thread cutting
6. Parting
7. Grooving
8. Drilling
9. Reaming
10. Taper turning
11. Step turning

1. TURNING:-

Turning is the process of getting rid of the extra material from the workpiece to generate a cylindrical surface to the required length. The workpiece is placed right between the centre and is spinned at the desired pace. The tool is fed either linearly in the direction parallel or perpendicular to the axis of rotation of the workpiece.

The primary motion of cutting in turning is the rotation of the workpiece, and the secondary motion of cutting is the feed motion.

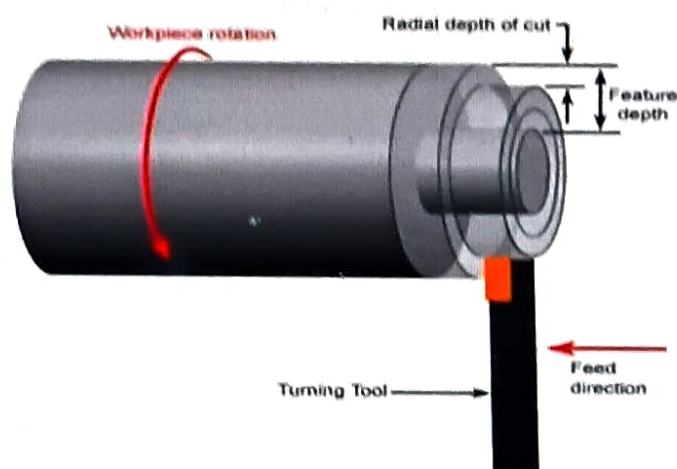
This operation is used to reduce the diameter of a part to a desired dimension for the desired length of job.



2. FACING:-

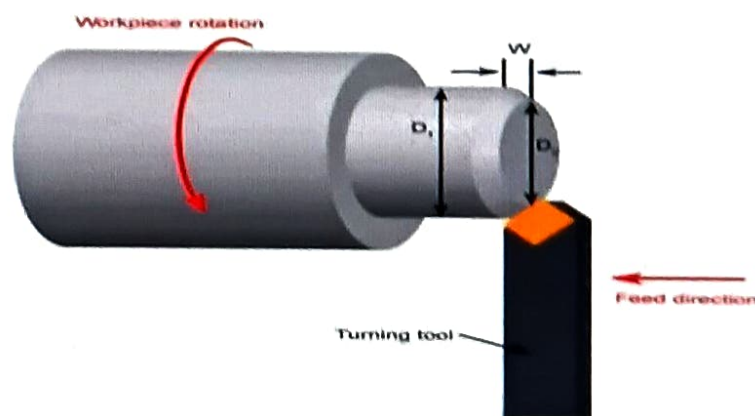
Facing is the operation of machining the end or face of the job. Facing is used for reducing the length of the workpiece by machining or cutting the end face of workpiece. Facing operation is

carried out by using a turning tool. The operation involves feeding the tool perpendicular to the axis of rotation of the workpiece from outer surface to centre.



3. CHAMFERING:-

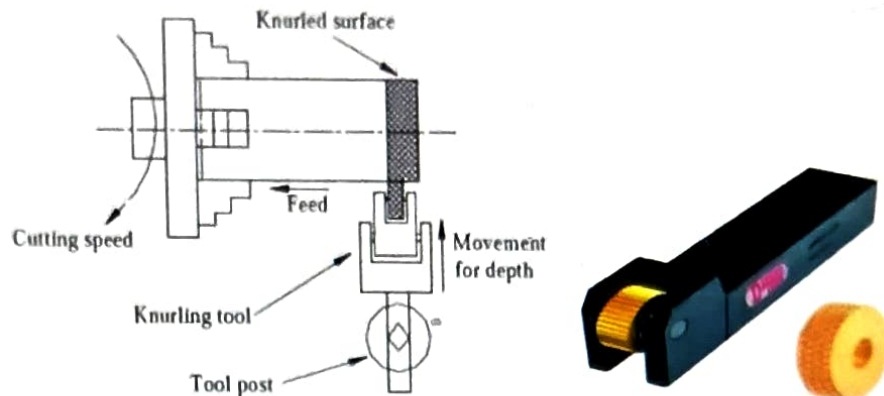
Chamfering is the operation of beveling the sharp ends of a workpiece to avoid any injuries to the persons using the finished product. Chamfering is similar to form turning and is done with a chamfering tool that has its cutting edge at the desired chamfer angle, usually 45° . Chamfer is provided for better look, to enable nut to start freely on threaded workpiece, to remove burrs as well as to protect the end of the workpiece from being damaged.



4. KNURLING OPERATION:-

The knurling operation is the process of attaining a diamond structure on the workpiece. It is done on certain components for effective gripping to prevent it from slipping while handling it, example is thumbscrews, etc.

The knurling tool consists of a set of hardened steel rollers in a holder with the teeth cut on their surface in a definite pattern. The tool is held rigidly on the tool post and the rollers are pressed against the revolving workpiece to squeeze the metal against the multiple edges.

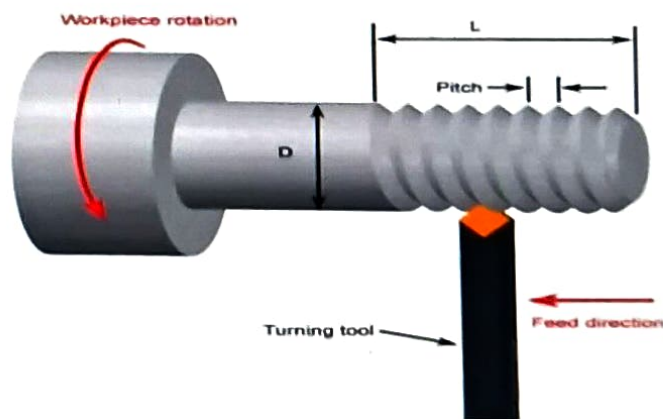


5. THREAD CUTTING OPERATION :-

Thread cutting is a type of operation in which the threads are being cut on the internal and the outer surface of the workpiece as per the requirement.

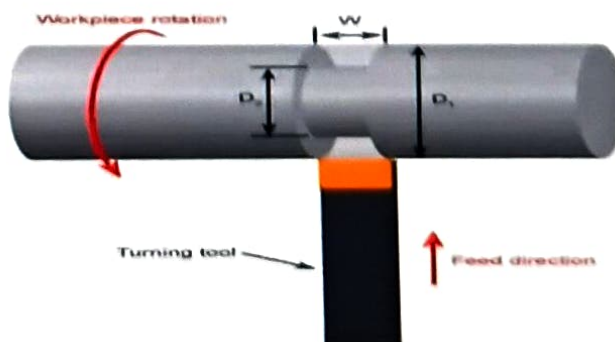
In the thread cutting operation only the automatic feed is given.

The automatic feed required for the thread cutting operation is given by using lead screw and the feed gear box.



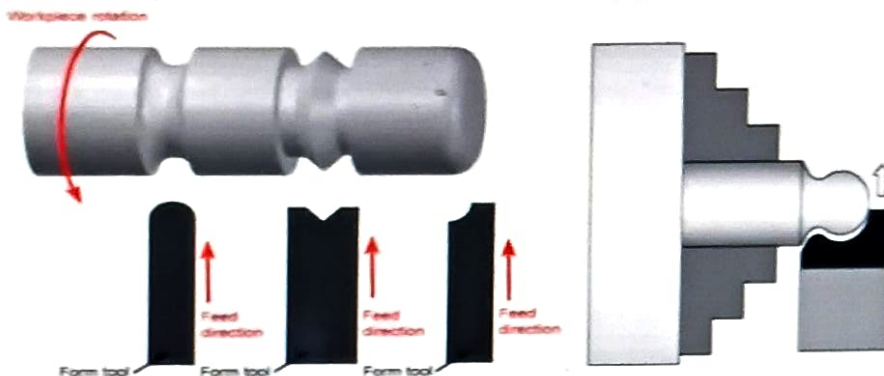
6. PARTING:-

Parting is operation in which the deep groves are being made on the parent material to remove the specific portion from the parent material resulting in dividing the workpiece into two or more parts.



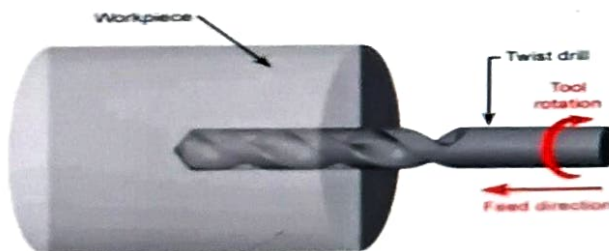
7. GROOVING :-

Grooving or groove cutting is the process of producing a narrow groove on the surface of a cylindrical job. The diameter of the workpiece is reduced slightly from the surface over a narrow width. Depending on the shape of the tool, it is possible to have square, round, beveled or any other shape grooves. In grooving operation, the shape of the tool is reproduced on the workpiece; hence, this process is also known as form turning operation.



8. DRILLING:-

Drilling is the process of creating a hole that is of a cylindrical shape in the wall. It can be done with the help of a rotating mechanism. The rotating part of the slicer better known as the drilling drill. During this procedure the workpiece is moving in and around a faceplate, or a chuck and the drill is held against the drill holder of the tailstock and the chuck of the drill chuck. The material is pulled out and is impacted by the motion on the tailstock axle. This technique is accepted widely for the drilling shaped workpiece.

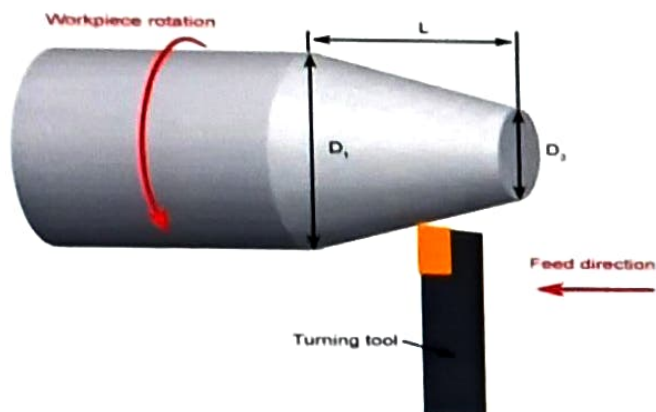


9. REAMING:-

Reaming is the process which could be finalizing and classifying a hole which already has been bored or drilled. The mechanism is known as reamer which comes fitted with multiple plate cutting fringes. The reamer is held up against a tailstock axle, directly or through the drill chuck and is uphold at a fixed destination while the work is revolved at a very slow pace.

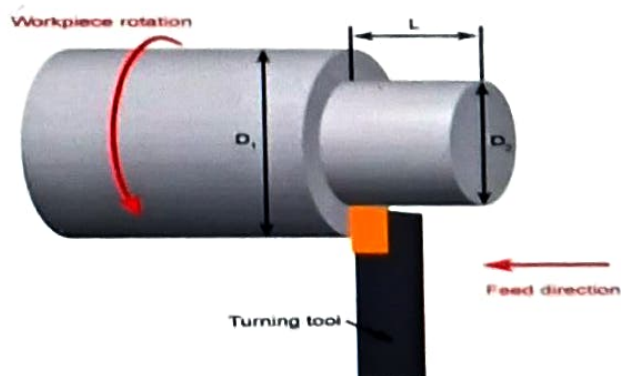
10. TAPER TURNING :-

This operation is performed to produce a conical surface of uniform reduction in diameter from a cylindrical workpiece.



11. STEP TURNING:-

In step turning, a series of steps with varying diameters is created on the workpiece, resulting in a stepped feature.



Prepare the Process plan as per the given drawing for Plain Turning, Step Turning, Knurling and Chamfering

AIM:

To perform Plain Turning, Step Turning, Chamfering, Thread cutting and Knurling operation on the given work piece.

MATERIAL REQUIRED:

Mild Steel rod of 25 mm diameter and 100 mm long

TOOLS REQUIRED:

Vernier calipers, steel rule, spanner, chuck spanner, and H.S.S. single point cutting tool, parting tool and V- cutting tool.

SPECIFICATION OF LATHE:

Length of bed 1390 mm

Width of bed 200 mm

Height of centers 165 mm

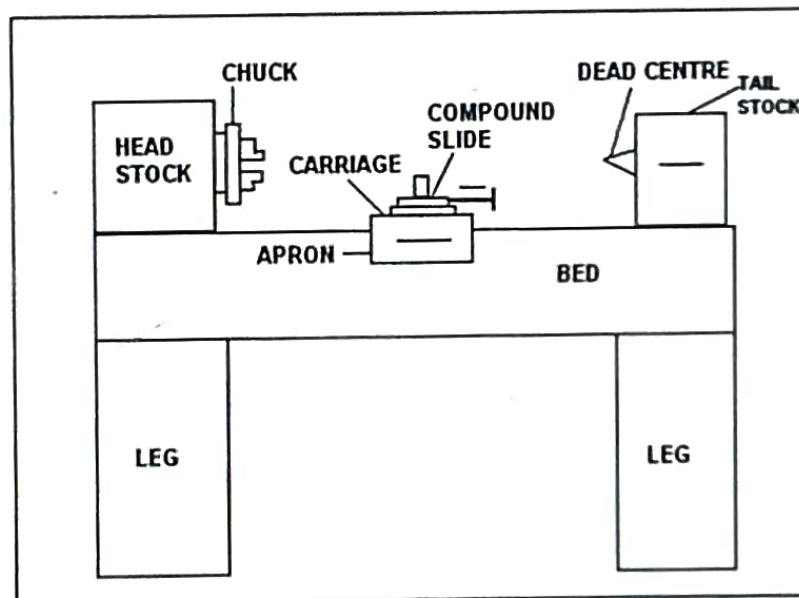
Admit between centers 700 mm

Lead screw pitch 4TPI

Power of the motor 1 H.P.

THEORY:

Lathe removes undesired material from a rotating work piece in the form of chips with the help of a tool which is traversed across the work and can be fed deep in work. The tool material should be harder than the work piece and the later help securely and rigidly on the machine. The tool may be given linear motion in any direction. A lathe is used principally to produce cylindrical surfaces and plane surfaces, at right angles to the axis of rotation. It can also produce tapers and bellows etc.



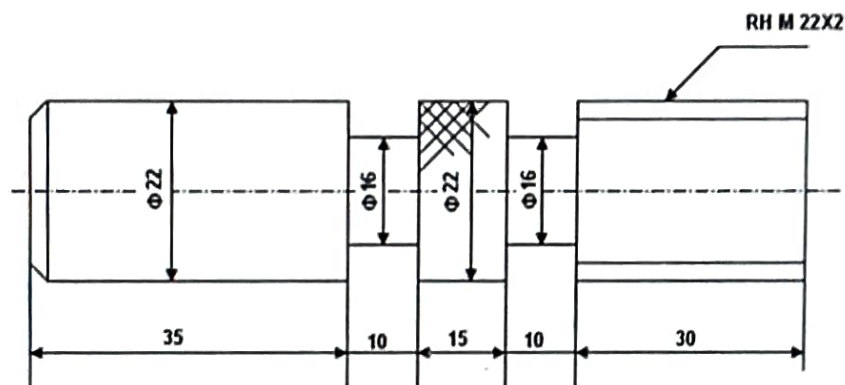
BLOCK DIAGRAM OF A LATHE

A lathe basically consists of a bed to provide support, a head stock, a cross slide to traverse the tool, a tool post mounted on the cross slide. The spindle is driven by a motor through a gear box to obtain a range of speeds. The carriage moves over the bed guide ways parallel to the work piece and the cross slide provides the transverse motion. A feed shaft and lead screw are also provided to power the carriage and for cutting the threads respectively.

SEQUENCE OF OPERATIONS:

- Centering
- Facing
- Plain turning
- Chamfering
- Step turning
- Grooving
- Thread cutting
- Knurling

PROCEDURE:



THREAD CUTTING AND KNURLING

ALL DIMENSIONS ARE IN MM

- The work piece is fixed in a 3 – jaw chuck with sufficient overhang.
- Adjust the machine to run the job to required cutting speed.
- Fix the cutting tool in the tool post and centering operation is performed so that the axis of the job coincides with the lathe axis.
- Facing is performed by giving longitudinal depth of cut and cross feed.
- Perform plain turning operation until the diameter of the work piece reduced to 20mm.
- Chamfering operation is done according to the given dimensions.

- Then reverse the work piece in the chuck and plain turning operation is performed according to the given dimensions.
- Using V-cutting tool and parting off tool perform grooving operation to the required dimensions.
- Reduce speed of the spindle by engaging back gear and use Tumbler
- feed reversing mechanism to transmit power through the lead screw.
- And calculate the change gears for the required pitch to be made on the work piece.
- Using half nut mechanism perform thread cutting operation(right hand threading) according to the given dimensions and continues it until required depth of cut is obtained.
- At the same speed knurling operation is performed using knurling tool.
- For every operation check the dimensions using vernier calipers.

PRECAUTIONS:

- Before starting the spindle by power, lathe spindle should be revolved by one revolution by hand to make it sure that no fouling is there.
- Tool should be properly ground, fixed at correct height and properly secured, and work also be firmly secured.
- Chips should not be allowed to wind around a revolving job and cleared as often as possible.
- Before operating threading operation, V-tool should be properly ground to the required helix angle.
- Apply cutting fluids to the tool and work piece property.
- No attempt should be made to clean the revolving job with cotton waste.
- On hearing unusual noise, machine should be stopped.

RESULT: The job is completed successfully and safely.